

TwinSync — 7-Minute Presentation Script

ASEAN GeoAI Fusion 2026 · HACK_MY_071 Three speakers · 7 minutes presentation · 3 minutes Q&A

Every number here is measured, not estimated. They come from `data/results.json`, regenerated by a controlled A/B run and asserted by 19 tests that fail the build if the README and the artifact disagree. Change the scenario → re-run the A/B → re-check this script before you speak.

This script is 1128 spoken words. At a rehearsed 165 words a minute that is about 410 seconds of speech, leaving ~10 seconds of deliberate pause inside the 7:00 slot. **Adding a sentence means cutting one**. Time it on the laptop before you trust it — being over-length is how a good pitch loses.

Speaker assignment

	Speaker	Segment	Clock	Spoken words
1	Kean Hong	The problem, and the five questions	0:00 – 1:50	307
2	Edrian	Live demonstration	1:50 – 5:00	469
3	Xu Zhe	What we measured, and what it cost	5:00 – 7:00	352

Speaker 2 holds the screen longest but speaks least — his 190-second demo carries about 30 seconds of deliberate silence while the captions land. Actual speaking time is roughly 112 / 171 / 128 seconds.

Rule: Speaker 2 narrates the demo end to end. No handoff mid-demo — the guided track runs on its own clock and a handoff is where timing breaks.

The spine: five questions

Five questions an operator must answer during an outage. Speaker 1 poses them, Speaker 2 answers each live, Speaker 3 prices them. **Say the question numbers out loud** — judges score against a rubric, so give them a checklist to tick.

#	The question	The method	Lands at
1	<i>Who actually lost signal?</i>	3D ray-cast over Copernicus GLO-30 + extruded OSM, 60 % first-Fresnel clearance	4:20
2	<i>How fast did we even find out?</i>	On-tower EWMA + Mahalanobis , confirmed by a 3 KB ONNX autoencoder	2:13
3	<i>Is this three problems, or one?</i>	ST-DBSCAN (Birant & Kut 2007) — 900 m / 600 s, plus a fault-family gate	2:31 + 3:18
4	<i>Which outage do we fix first?</i>	Deterministic reach × severity × criticality × urgency , quadratic SLA term	4:47
5	<i>Does the weather change the answer?</i>	ITU-R P.838 rain fade + DEM flood mask + A* re-route on the OSM graph	3:52

If a judge wants one more level of detail, the file is the answer: `twinsync/coverage.py`, `edge/detector.py`, `twinsync/stdbscan.py`, `twinsync/priority.py`, `twinsync/routing.py`. Nothing in that table is a library call we wrapped — the clustering, the Fresnel test and the dispatcher are ours.

Before you begin

```
python -m uvicorn twinsync.server:app --port 8000
python scripts/verify_ui.py http://127.0.0.1:8000 shots/
```

The dashboard fails silently — blank WebGL canvas, clean console, healthy-looking HUD. `verify_ui.py` is the only thing that catches it.

⚠ **It reports one known false alarm:** `storm click produced no cells in state`. The check reads state before the next simulation step computes the cells; it fails the same way on untouched code. Any **other** error is real — do not go on stage with a second one.

Open `http://localhost:8000` full-screen. Do **not** press `D` yet — Speaker 2 presses it at 1:50. Fallback video on the desktop and the USB stick.

SPEAKER 1 — Kean Hong

0:00 – 1:50 · 307 spoken words

0:00 – 0:12 · Opening

Slide: title.

Good afternoon. We're team HACK_MY_071. TwinSync is a live three-dimensional twin of a telecom network, over the Kuala Lumpur CBD.

0:12 – 0:45 · How this works today

Slide: the "today" sequence.

Here's how a network fault is handled today.

A site fails. Nobody knows — until a customer complains, ten minutes later.

The ops centre sizes it with a flat circle around the tower. Tickets get worked in arrival order, so a transport interchange waits behind an empty car park. And the crew is whoever looks closest in metres.

If it's raining, none of that is still true when the van arrives.

0:45 – 1:22 · The core problem

Slide: the compare screenshot. Left calm, right red.

Here's the sentence this whole project is built on.

A two-dimensional map cannot tell you who lost signal, because signal is blocked in three dimensions. That's why crews get sent to the wrong place.

One outage, two ways. Left: the flat model dispatch uses today. Right: true line of sight — ray-cast against the elevation model and every building at its real height, requiring **sixty per cent first-Fresnel clearance**, the standard radio planning rule. Same outage, same instant.

The flat map says **three** buildings are dark — it decides the next cell has the rest. It's wrong: a two-hundred-metre tower is in the way. True line of sight says **forty-seven buildings. Nine thousand and twenty-six subscribers.**

Nine thousand and twenty-three people are off the air — and the tool the operator is holding is quietly confident they're fine.

Pause. One full beat.

1:22 – 1:42 · The five questions

And that's only question one of five.

One — who actually lost signal? Two — how fast did we even find out? Three — is this three problems, or one storm? Four — which one do we fix first? Five — does the weather change the answer?

Four real sources answer all five: Copernicus elevation, OpenStreetMap, Sentinel-2, and ITU-R rain physics. Offline, on this laptop.

1:42 – 1:50 · Handoff

You'll watch all five answered live, in three minutes. **Edrian will show you.**

SPEAKER 2 — Edrian

1:50 – 5:00 · 469 spoken words

Press D as you start speaking. Fifteen beats, three minutes ten, fires on its own. You narrate *over* it. Times are pitch clock — if you drift, trust the screen, not this page.

1:50 – 2:05 · The city

The KL business district as our twin sees it — not a flat map, a city with height.

Notice what the network *isn't* sending: that uplink figure past ninety-eight per cent never leaves the tower.

2:13 · QUESTION TWO — how fast did we find out?

KL-03 goes amber. MTTD tile lights green.

First fault — a rooftop amplifier cooking.

Question two: how fast did we find out? The detection tile says **two point eight seconds**.

Today that's a complaint, ten minutes later.

That's running on the tower: an **EWMA baseline and a Mahalanobis distance** at full sample rate, confirmed by a **three-kilobyte ONNX autoencoder**.

2:20 — splits to Compare. Let it land, and say nothing.

2:31 · QUESTION THREE — one problem, or three?

KL-13 fails. Cluster CL-001.

Second site down, two hundred and ninety metres away.

Question three: second problem, or the same one? This is **ST-DBSCAN** — two radii, nine hundred metres and ten minutes, plus a fault-family check. These two are three hundred metres and two minutes apart, so it's **one** incident, and the second job folds into the trip already running.

2:43 — truck-roll tile holds at 1.

That saves a truck roll. It also makes the second job wait — we report that too.

3:06 · The monsoon

A convective cell crosses the city — rain fade on the eighteen-gigahertz backhaul, integrated along the hop from **ITU-R P.838**.

3:18 · QUESTION THREE, the other half

KL-09 fails, tagged *ISOLATED*.

Third fault — and this time it says **isolated**. Twelve hundred metres away, different failure type. Not "nearby means the same" — nearby, recent, **and** a compatible fault. This one fails two of the three. The system can say no.

3:52 · QUESTION FIVE — does the weather change the answer?

Cyan flooded segments. Routes redraw.

Question five — my favourite.

The elevation model says which roads sit lowest. The storm says where rain is falling. The OSM graph says which are on a crew's route.

Watch the cyan — those roads just went under. Travel time reprices, **A-star re-plans over the street graph**, and **a different crew wins the job**.

4:14 · Four sites dark

A fourth site loses mains. Now look at the comparison.

4:20 · QUESTION ONE — who actually lost signal?

Compare. Verdict bar.

Question one. Who actually lost signal?

Four sites down. The flat model reports **three buildings**. True line of sight reports **forty-seven, and nine thousand and twenty-six subscribers**.

And it isn't raising a false alarm — it isn't alarming at all. **It's reassuring, and that's worse.**

Nine thousand and twenty-three people it's quietly confident are fine.

Pause two seconds. This is the peak of the pitch.

4:28 · The last van rolls

A fifth site, small and far away. The last van rolls — **every crew is committed**.

4:47 · QUESTION FOUR — which one do we fix first?

KL-04

KL-04 dark. A crew is pulled off mid-route.

And now the one that changes the day. Forty-two thousand subscribers, two pieces of critical infrastructure. **Question four: which do we fix first?**

We score it: reach, times severity, times criticality, times time waited — and that last term is **quadratic**, so a small job never starves. This scores **a hundred and forty-four**. Everything else is under five.

Every van is busy — so the dispatcher **preempts**, pulling a crew off a smaller job mid-route. And it chose the crew **further away**, because the road was faster.

4:58 – 5:00 · Handoff

The gap doesn't close as things get worse — it widens.

Xu Zhe will tell you what we measured, including the number that got worse.

SPEAKER 3 — Xu Zhe

5:00 – 7:00 · 352 spoken words

5:00 – 5:38 · What we measured

Slide: the A/B table.

A controlled experiment: same scenario, same seed, same world, same four crews, run twice. Only the dispatch logic changes.

Detection: ten minutes, down to under a second.

Restore time: forty point three minutes, down to twenty-eight point nine — twenty-eight per cent, almost all of it detection, because ten minutes waiting for a phone call is ten minutes a crew isn't driving.

Our ninetieth percentile improves too. **Subscriber-minutes lost: down four hundred and forty-five thousand.** Uplink falls ninety-eight and a half per cent.

5:38 – 6:20 · The number that got worse

Same slide. Do not rush this — it is the most persuasive part of the pitch.

Now the part I want to tell you about, because you'd find it anyway.

Our truck rolls did not go down. Six, and six. And we drive three point eight kilometres further.

Batching saves a roll. Preemption spends it straight back — pull a crew off a job, and that job needs its own trip later. That's the U-turn.

So honestly: **we spend one van-trip and four kilometres to restore a fifty-three-thousand subscriber site sooner.** An operator may take that deal — but it's a trade, not a saving.

An earlier version of this slide claimed one point one two million ringgit. It came from a fleet never busy enough to preempt — so the preemption we'd been describing had **never once executed.** We fixed it, and the money went to zero.

6:20 – 6:40 · What it is worth

Slide: the projection.

So where's the value? Restored service. Two thousand sites at four faults a year: **nineteen point eight million subscriber-hours.** Those multipliers are the weakest part — so we made them clickable.

Click the ROI tile.

Five thousand sites — **forty-nine point four million**. Only the scaling is assumed.

6:40 – 6:52 · Why you can trust it

Every model artifact ships with a card and a live SHA-256 hash. **Two hundred and twenty-two tests** on every push — nineteen fail the build if our README stops matching the run.

And the machine learning never dispatches anything — the dispatcher is deterministic and auditable. The risk model is advisory.

6:52 – 7:00 · Close

Three numbers.

Nine thousand and twenty-three people a flat map says are fine, and who aren't.

Ten minutes, to under three seconds, to find out.

Nineteen point eight million subscriber-hours restored a year — every assumption on screen, and clickable.

Thank you.

Stop. Add nothing after "thank you".

Q&A — 3 minutes

Routing: Edrian takes technical. Xu Zhe takes numbers and methodology. Kean Hong takes problem, market, next steps.
Answer in under thirty seconds and stop.

△ "Which hospital?" / if anyone says *hospital*

We don't claim one — and don't say it on stage. The critical assets are MRT stations — Bukit Bintang, TRX, Hang Tuah — a transit interchange, and a police post. Twelve critical buildings in the area of interest, tagged from OpenStreetMap.

"So it saves no money?"

In fuel and truck rolls on this scenario — correct, zero. Batching saves a roll and preemption spends it back. The measured win is service restoration: four hundred and forty-five thousand subscriber-minutes in one simulated hour. If your cost model prices an SLA breach or a churned subscriber, that converts. Ours doesn't, because we'd be inventing the conversion rate.

"Your fuel and distance got worse."

Yes — three point eight kilometres, and we left it in the table. That's the cost of preemption: pulling a crew off a job means the abandoned job needs a second trip. We could hide it by not preempting — but then the biggest outage waits behind the smallest, which is the problem we set out to fix.

"Isn't this all synthetic?"

Partly, and the README says which parts. Four real sources: Copernicus elevation, OpenStreetMap, a named Sentinel-2 scene, ITU-R rain physics. The *failure labels* are synthetic because no public telco failure dataset exists. It's also why the model advises and a deterministic dispatcher decides.

"Only twenty-eight per cent on MTTR?"

On-site repair dominates and is identical in both arms — inflating it would mean modelling a repair that goes faster for no reason. Twenty-eight per cent is almost entirely detection. Our p90 improves as well, which it didn't in an earlier configuration, and we reported that one too.

"Your tile says 2.8 seconds, your table says under one."

Different populations, both in the README. The table pairs MTDD with MTTR, so it counts only incidents that ran all the way to repair — that's zero point seven seconds. The dashboard tile averages *every* fault seen, so it matches the log on screen. Per fault: point six, point six, point eight, one point two, two point eight, five point eight.

"How does this generalise beyond KL?"

The pipeline is driven by coordinates, not tuned to this city. Our portability test confirms Copernicus tile naming resolves for Jakarta, Bangkok, Manila, Hanoi, Singapore and Phnom Penh — including the southern-hemisphere case a naive integer cast gets wrong by a hundred kilometres. A second city still needs the data bake and a scripted timeline.

"Does it scale past fifteen towers?"

Runtime is linear in sites — at ninety sites we use three and a half per cent of the frame budget, roughly seven hundred sites in one process. Our clustering builds a full neighbour matrix instead of a spatial index, so it's quadratic. Two milliseconds at ninety sites, but that's the wall, and a spatial index is the fix.

"How accurate are your building heights?"

Mean absolute error thirty-six metres, median twenty-two, cross-validated on held-out CBD buildings. That's weak and we say so. The production answer is LiDAR. Imputed buildings are tinted differently so a guess never reads as a survey.

"Why are sites still down when the run ends?"

The window is one hour; the last fault lands at twenty-three minutes with a thirty-five minute repair. Unrepaired outages accrue subscriber-minutes in **both** arms — we fixed that specifically, because counting only repaired incidents rewards an arm for leaving work undone.

"What would six more months buy?"

A real asset register, to replace the weakest input to our risk model. Temporal validation — holding out weeks forward, not just sites. A spatial index for the clustering. And a second ASEAN city bake.

Rehearsal notes

1. **Rehearse five times against a timer**, on the presentation laptop, at venue brightness. Speaker 2 is fixed at 195 seconds by the demo clock — all flexibility is in segments 1 and 3.
2. **Say the question numbers out loud.** "Question two." "Question four." It is the only reason the demo is legible to someone seeing it once.
3. **The handoff lines are load-bearing.** "Edrian will show you." and "Xu Zhe will tell you what we measured, including the number that got worse." The second sets up your strongest thirty seconds.
4. **Speaker 1 finishes by 1:50.** Running over? Cut the rain sentence, then compress the four sources to "four real geospatial sources".
5. **Speaker 3 reaches the close by 6:52.** Behind? Cut the trust paragraph. **Never cut "the number that got worse"** — it is worth more than the trust paragraph and the projection combined.
6. **Nobody speaks over the demo captions.** They fire on their own. Pause and let them land.
7. **The 47-versus-3 moment is a 100-second window** (demo $t=1150 \rightarrow t=1250$). Drift late and the screen reads 134 versus 71 — still a gap, less brutal. Say the nine-thousand number at 4:20.
8. **If the demo fails:** "Our dashboard isn't cooperating, so let me show you the recorded run." Play the fallback, narrate the same script. Do not troubleshoot on stage.
9. **Off-script keys for Q&A:** **F** fails the selected tower, **S** injects a storm, **W** raises the water, **3** jumps to Compare. All work live if a judge asks to see something again.
10. **If Edrian is running long, jump.** **Next** `>` on the caption card (or `→` / PageDown, including from a clicker) moves the scenario clock straight to the next beat — screen, queue and log with it — so a segment can be recovered without cutting the close. **< Prev** goes back. Needs `python scripts/bake_checkpoints.py` run beforehand; the card says so if it was not.